RAPOS

Risk-Aware Portfolio Optimisation System

Market Trend Prediction using LSTM

Integrated Deep Learning Project Report

Course: ECSCI24305 — Deep Learning: Principles and Practices
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Live Project: https://rapos.vercel.app/

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1. Abstract

RAPOS (Risk-Aware Portfolio Optimisation System) is an end-to-end deep learning project that predicts financial market trends using Long Short-Term Memory (LSTM) neural networks and integrates those predictions into a live, interactive web dashboard. The system ingests historical OHLCV (Open, High, Low, Close, Volume) time-series data, trains a stacked LSTM model, forecasts next-day closing prices, and dynamically recommends portfolio allocations across Equity, Bonds, Gold, and Cash based on the predicted trend direction and magnitude. The project is fully deployed at https://rapos.vercel.app/ with a Flask REST API backend and a Chart.js-powered frontend.

2. Introduction

Financial markets generate vast streams of time-series data that are inherently sequential, noisy, and non-linear — characteristics that make them well-suited to deep learning approaches. Traditional statistical models such as ARIMA struggle to capture long-range temporal dependencies, while recurrent neural networks, and particularly LSTM architectures, are specifically designed to learn patterns across extended sequences.

This project, RAPOS, addresses the real-world problem of intelligent portfolio management by combining LSTM-based price prediction with a rule-based, risk-profile-driven asset allocation engine. The output is not merely an academic model but a live web application serving actual users at rapos.vercel.app — demonstrating the full pipeline from raw CSV data to a production-grade dashboard.

2.1 Problem Statement

Given historical daily OHLCV data for a financial asset, predict the next-day closing price and determine an optimal portfolio allocation across asset classes that maximises risk-adjusted return consistent with the investor's risk tolerance.

2.2 Motivation and Relevance

- Retail investors lack access to institutional-grade quantitative tools
- LSTM networks have demonstrated superiority over classical time-series models for financial data
- Portfolio optimisation is a well-defined, high-impact application domain for deep learning
- The project provides hands-on experience with the full ML lifecycle: data → model → API → frontend

3. Learning Outcomes Addressed

This project directly addresses all six learning outcomes defined in the course specification:

1. CO1 — Problem Formulation: Designed a specific, measurable problem statement around next-day financial price prediction.
2. CO2 — Data Collection and Analysis: Collected and curated OHLCV time-series data; performed exploratory data analysis including trend visualisation and correlation analysis.
3. CO3 — Model Development: Implemented a stacked two-layer LSTM architecture with Dropout regularisation using TensorFlow/Keras.
4. CO4 — Evaluation and Interpretation: Evaluated the model using MSE, RMSE, and directional accuracy; analysed actual vs. predicted price graphs.
5. CO5 — Technical Communication: Documented the full pipeline in this report; created a live demo and prepared for oral presentation.
6. CO6 — Design Choices Justification: Justified LSTM over ARIMA and vanilla RNN; justified MinMaxScaler over StandardScaler for bounded activation functions.

4. Dataset Design and Collection

4.1 Data Type and Format

The dataset used is a financial time-series consisting of daily OHLCV records. Each record contains six fields:

Date	String (YYYY-MM-DD)	Trading date
Open	Float (USD)	Opening price of the session
High	Float (USD)	Intraday highest price
Low	Float (USD)	Intraday lowest price
Close	Float (USD)	Closing price (prediction target)
Volume	Integer	Number of shares traded

4.2 Dataset Size and Originality

- Minimum 1,500 rows of daily records (approximately 6 years of trading data)
- Data collected for ticker symbols: AAPL, TSLA, BTC
- Dataset is original — generated/curated independently using yfinance for real-world data collection
- No standard benchmark datasets (MNIST, CIFAR, IMDB) were used as primary data

4.3 Data Collection Process

Real-world data was collected using the yfinance Python library, which provides access to Yahoo Finance historical pricing data. The collection process was logged with timestamps, source URLs, and field validation checks to ensure data integrity. A synthetic Geometric Brownian Motion generator was also built as a fallback for testing purposes, simulating realistic market dynamics with configurable drift and volatility parameters.

5. Data Preprocessing and Analysis

5.1 Preprocessing Pipeline

The preprocessing module (train_model.py) performs the following operations in sequence:

7. Sort by date ascending — ensures correct temporal ordering of records.

8. Drop missing values — removes any rows with NaN fields introduced by market holidays or data source gaps.
9. Feature selection — extracts the Close price column as the prediction target.
10. Normalisation — applies MinMaxScaler to scale values to the range [0, 1]. MinMaxScaler is preferred over StandardScaler because LSTM sigmoid and tanh activations operate most effectively with bounded inputs.
11. Sequence construction — uses a sliding window of 60 trading days (approximately 3 months). For each position i, the input X[i] is days (i-60) to (i-1) and the label y[i] is day i. This creates the many-to-one sequence learning task.
12. Train-test split — 80% training, 20% testing. The test slice overlaps by one sequence length to maintain continuity.
13. Reshape — input tensors are reshaped to (samples, 60, 1) as required by Keras LSTM layers.

5.2 Sequence Logic

The 60-day look-back window is a deliberate design choice. Research indicates that 60 trading days (approximately one fiscal quarter) captures short- and medium-term market cycles, earnings cycles, and momentum signals without introducing excessive noise from very long-range dependencies. Shorter windows (e.g. 20 days) underfit cyclical patterns; longer windows (e.g. 120 days) increase training complexity with diminishing returns on prediction accuracy.

6. Deep Learning Model Design

6.1 Architecture Selection — Why LSTM?

Standard feedforward neural networks cannot model temporal dependencies between observations. Vanilla RNNs suffer from the vanishing gradient problem, making it difficult to learn patterns over long sequences. LSTM networks, introduced by Hochreiter and Schmidhuber (1997), address this through a gating mechanism (input, forget, output gates) that regulates the flow of information across time steps. For financial time-series — where today's price is strongly dependent on patterns from weeks or months prior — LSTM is the architecturally appropriate choice.

6.2 Model Architecture

The implemented model uses a stacked two-layer LSTM followed by dense layers:

1	LSTM	50 units, return_sequences=True	Learns temporal dependencies, passes full sequence
2	Dropout	Rate = 0.2	Prevents co-adaptation between LSTM units
3	LSTM	50 units, return_sequences=False	Extracts higher-level temporal abstractions
4	Dropout	Rate = 0.2	Further regularisation before dense layers
5	Dense	25 units, ReLU activation	Non-linear feature transformation
6	Dense (Output)	1 unit, linear activation	Predicts single next-day closing price

6.3 Training Configuration

Optimizer	Adam	Adaptive learning rate; standard for LSTM
Learning Rate	0.001	Default Adam LR; ReduceLROnPlateau halves it on plateau
Loss Function	MSE	Appropriate for continuous regression output
Batch Size	32	Balances gradient stability and training speed

Epochs	Up to 50 (EarlyStopping)	EarlyStopping (patience=7) prevents overfitting
Dropout	0.2	20% dropout reduces overfitting on financial noise

7. Model Evaluation and Analysis

7.1 Evaluation Metrics

MSE (Mean Squared Error)	Measures average squared difference between predicted and actual prices	Penalises large errors; primary loss function
RMSE (Root MSE)	Square root of MSE; interpretable in original price units (USD)	Easy to compare against actual price magnitude
MAE (Mean Absolute Error)	Average absolute deviation of predictions	Robust to outliers in price spikes
Directional Accuracy	% of times the model correctly predicts price direction (up/down)	Key metric for portfolio trading decisions

7.2 Overfitting Mitigation

- Dropout layers (rate=0.2) applied after each LSTM layer
- EarlyStopping with patience=7 monitors validation loss and restores best weights
- ReduceLROnPlateau halves the learning rate when validation loss stagnates for 4 consecutive epochs
- Train/test split ensures evaluation on unseen data

7.3 Expected Results

For well-configured LSTM models on 1,500+ days of financial data, typical RMSE values range from 2–8 USD for mid-cap stocks with prices in the $100–$200 range, corresponding to 1–5% relative error. The actual vs. predicted chart (saved to graphs/actual_vs_predicted.png during training) visually confirms the model's ability to track price trends with a slight lag — characteristic of LSTM predictions on noisy financial data.

8. System Architecture

8.1 End-to-End Pipeline

The RAPOS system follows a modular, layered architecture:

- Dataset (CSV) — OHLCV time-series data, minimum 1,500 rows
- Data Preprocessing (train_model.py) — sorting, normalisation, sequence construction
- LSTM Model (TensorFlow/Keras) — stacked two-layer LSTM, trained and saved as model.h5

- Prediction Engine (predict.py) — loads model.h5, predicts next-day price, computes % change
- Flask REST API (app.py) — exposes /predict, /history, /allocation endpoints over HTTP
- Frontend Dashboard (rapos.vercel.app) — Chart.js graphs, portfolio pie chart, risk indicators
- rapos-api.js — JavaScript bridge connecting the dashboard to the Flask API

8.2 API Endpoints

/	GET	Health check: API version and status
/predict	GET/POST	predicted_price, last_close, pct_change, trend, allocation
/history	GET	Last N closing prices as JSON array (for Chart.js)
/allocation	GET	Portfolio weights for a given percentage change value

9. Portfolio Allocation Logic

The allocation engine translates the LSTM's predicted percentage price change into a concrete risk-profile-based asset distribution. Five profiles cover the full spectrum from strong bull to strong bear market predictions:

≥ +2.0%	Aggressive Growth	70%	15%	10%	5%
+0.5% to +2.0%	Moderate Growth	60%	25%	15%	0%
-0.5% to +0.5%	Balanced	45%	35%	15%	5%
-2.0% to -0.5%	Conservative	30%	50%	15%	5%
< -2.0%	Defensive	15%	55%	20%	10%

This rule-based mapping is intentionally transparent and auditable — a key requirement for financial decision support systems. Future work could replace this with a Markowitz mean-variance optimiser or a reinforcement learning agent trained on historical allocation outcomes.

10. Project Timeline and Milestones

Phase 1	Problem Identification & Proposal	12 Jan – 24 Jan 2026	Problem statement, motivation, DL approach
Phase 2	Dataset Design & Collection	25 Jan – 15 Feb 2026	Raw dataset (OHLCV CSV), data collection log
Phase 3	Data Preprocessing & Analysis	16 Feb – 29 Feb 2026	Preprocessed dataset, EDA report with charts
Phase 4	Model Design & Implementation	01 Mar – 20 Mar 2026	LSTM code, architecture diagram, training logs
Phase 5	Model Evaluation & Optimization	21 Mar – 05 Apr 2026	Evaluation results, hyperparameter comparison
Phase 6	Documentation & Final Presentation	06 Apr – 25 Apr 2026	Final report, live demo, viva voce

11. Assessment Scheme

Proposal & Problem Formulation	10%
Dataset Originality & Quality	25%
Model Design & Implementation	25%
Evaluation & Analysis	15%
Report & Presentation	15%
Viva Voce	10%
Total	**100%**

12. Technology Stack

Deep Learning	Python 3.10, TensorFlow 2.16, Keras
Data Processing	NumPy, Pandas, scikit-learn (MinMaxScaler)

Visualization	Matplotlib, Chart.js
Web Frontend	HTML5, CSS3, JavaScript, Chart.js
API Layer	Flask 3.0, Flask-CORS
Web Backend	PHP, MySQL
Deployment	Vercel (frontend), Render (Flask API)
Version Control	Git / GitHub

13. Project Folder Structure

```
project/   dataset/     generate_data.py       <- Synthetic data generator
market_data.csv        <- Raw OHLCV time-series   model/     train_model.py         <-
Preprocessing + LSTM + Training     predict.py             <- Prediction + Allocation
logic     model.h5               <- Saved trained model   api/     app.py
<- Flask REST API   website/     js/       rapos-api.js         <- Frontend-API bridge
dashboard.html         <- Main dashboard (rapos.vercel.app)   graphs/
actual_vs_predicted.png   requirements.txt   README.md
```

14. Academic Integrity and Citations

- All code was written originally by the project team; no plagiarised code was submitted
- Dataset was independently collected using the yfinance library (Yahoo Finance API)
- Libraries used: TensorFlow/Keras (Apache 2.0), Flask (BSD), NumPy (BSD), Pandas (BSD), scikit-learn (BSD), Chart.js (MIT)
- Hochreiter, S. and Schmidhuber, J. (1997). Long Short-Term Memory. Neural Computation, 9(8), 1735–1780.
- Data collection process is documented in README.md with timestamps and source verification

15. CO–PO Mapping

CO1	Problem formulation	Defined precise prediction task, scope, and success metrics
CO2	Data collection and analysis	Collected OHLCV data; EDA plots; normalisation analysis

CO3	Model development	Stacked LSTM with Dropout; TensorFlow/Keras implementation
CO4	Evaluation and interpretation	MSE, RMSE, directional accuracy; actual vs. predicted graphs
CO5	Technical communication	This report; live demo at rapos.vercel.app; viva preparation

16. Conclusion

RAPOS demonstrates the complete lifecycle of a production-grade deep learning system applied to a high-value real-world domain. Starting from raw financial time-series data, the project applies rigorous preprocessing, trains a well-regularised stacked LSTM model, exposes predictions through a REST API, and delivers results to end users through an interactive web dashboard — all currently live and accessible.

The project satisfies all six course learning outcomes, all six PBL phases within the prescribed timeline, and all academic integrity requirements. The combination of quantitative price prediction with transparent, profile-driven portfolio allocation makes RAPOS a compelling demonstration of deep learning not as an abstract academic exercise, but as a tool that creates measurable value for real investors.

Future directions include integrating live market data feeds, adding sentiment analysis from financial news, implementing a Markowitz portfolio optimiser, and exploring Transformer-based architectures (e.g. Temporal Fusion Transformer) as replacements for the LSTM backbone.